

NATIONAL UNIVERSITY OF SINGAPORE

NATIONAL UNIVERSITY OF SINGAPORE
READING ASSESSMENT 1 FOR
Semester 1 AY2025/26

CS1101S Programming Methodology

2025

Time Allowed: 45 minutes

INSTRUCTIONS TO CANDIDATES

1. This question paper contains 18 questions and comprises 10 printed pages, including this page.
2. You are also provided with an answer sheet to write your answers. It comprises 2 printed pages.
3. Use a pen or pencil to write your Student Number in the designated space on the front page of the answer sheet, and shade the corresponding circle completely in the grid for each digit or letter. **DO NOT WRITE YOUR NAME.**
4. You must submit only the answer sheet and no other documents.
5. All questions must be answered in the space provided in the answer sheet; no extra sheets will be accepted as answers.
6. For multiple choice questions, shade in the circle of the correct answer completely.
7. Each question has one correct answer. One mark is awarded for each correct answer and there is no penalty for a wrong answer.
8. The full score of this assessment is **17 marks**.
9. Answer all questions.
10. This is a **closed-book** assessment.

1. What is the alphabet letter that is in the box at the top-right corner on the front page of this question paper? Important: Please make sure your answer is correct because it determines how your answers to all the subsequent questions will be graded.

- (a.) A
- (b.) B
- (c.) C
- (d.) D

Section A

2. What kind of process does the following function give rise to for any integer argument $a > 1$?

```
def fun(a):  
    return a if a < 0 else fun(a - 2)
```

- (a.) An iterative process
 - (b.) A process that is neither iterative nor recursive
 - (c.) A substitution process
 - (d.) A recursive process
 - (e.) An infinite process
3. What kind of process does the following function give rise to for any integer argument $x > 1$?

```
def fun(x):  
    return 0 if x <= 1 else fun(1 + fun(x / 2))
```

- (a.) An iterative process
- (b.) A process that is neither iterative nor recursive
- (c.) A substitution process
- (d.) A recursive process
- (e.) An infinite process

4. What kind of process does the following function give rise to when applied to any integer argument $n > 0$ as a and fun as f ?

```
def fun(a, f):  
    return a if a == 0 else f(a - 1, fun)  
  
fun(n, fun)
```

- (a.) An iterative process
 - (b.) A process that is neither iterative nor recursive
 - (c.) A substitution process
 - (d.) A recursive process
 - (e.) An infinite process
5. What kind of process does the following function give rise to for any integer argument $n > 1$ and $m > 1$?

```
def fun(n, m):  
    return (  
        0  
        if n == 1  
        else fun(n - 1, m) + fun(m, n - 1)  
    )
```

- (a.) An iterative process
- (b.) A process that is neither iterative nor recursive
- (c.) A substitution process
- (d.) A recursive process
- (e.) An infinite process

6. What kind of process does the function `fun` give rise to for any integer argument $x > 0$ and integer argument $y > x$?

```
def fun(x, y):  
    return (  
        1  
        if x > y  
        else fun(x, y - 1) * 1  
        if x < y  
        else fun(x, y - 1)  
    )
```

- (a.) An iterative process
- (b.) A process that is neither iterative nor recursive
- (c.) A substitution process
- (d.) A recursive process
- (e.) An infinite process

Section B

7. Assuming x, y , and z are mutually distinct numbers, what number does the following expression evaluate to?

```
(z if y > z else y) if x > y else (z if x > z else x)
```

- (a.) The smallest of the three
- (b.) The largest of the three
- (c.) The median of the three
- (d.) The average of the three
- (e.) An error occurs during the evaluation.

8. Consider the following program.

```
def p(b, x, y):  
    return x if b else y  
  
def f(n):  
    return 1 if n == 1 else n * f(n - 1)  
  
p(True, 2 * 3, f(4))
```

How many multiplications are executed during the evaluation of this program if normal order reduction is used?

- (a.) 0
 - (b.) 1
 - (c.) 2
 - (d.) 3
 - (e.) 4
9. Consider the following program.

```
def p(b, x, y):  
    return x if b else y  
  
def f(n):  
    return 1 if n == 1 else n * f(n - 1)  
  
p(True, 2 * 3, f(4))
```

How many multiplications are executed during the evaluation of this program if applicative order reduction is used?

- (a.) 0
- (b.) 1
- (c.) 2
- (d.) 3
- (e.) 4

10. The primitive function `print` prints the argument to which it is applied and returns `None`. `D`, defined below, prints its argument and returns that same argument. Consider the following program.

```
def D(x):  
    print(x)  
    return x  
  
def d(x):  
    return 1 + D(D(x) + D(2))  
  
print(d(1))
```

In what order are the numbers 1, 2, 3, 4 printed?

- (a.) 1 2 3 4
- (b.) 4 3 2 1
- (c.) 2 1 4 3
- (d.) 4 3 1 2
- (e.) 4 2 3 1

Section C

11. We specify that the function `AM`, when applied to any two numbers `x` and `y`, should return their arithmetic mean.

```
def AM(x, y):  
    return x + y / 2
```

Which one of the following statements is correct?

- (a.) The function `AM` does not meet the specification because it causes a run-time error.
- (b.) The function `AM` meets the specification.
- (c.) The function `AM` does not meet the specification because it has syntax error(s).
- (d.) The function `AM` does not meet the specification because it returns incorrect results for all valid argument values.
- (e.) The function `AM` does not meet the specification because it returns correct results for some but not all valid argument values.

12. We specify that the function `digit_sum`, when applied to any positive integer argument `n`, should return the sum of its digits.

```
def digit_sum(n):  
    return (  
        n  
        if n < 10  
        else (lambda d: d + digit_sum(math.floor(n / 10)))(n % 10)  
    )
```

Which one of the following statements is correct?

- (a.) The function `digit_sum` does not meet the specification because it causes a run-time error.
 - (b.) The function `digit_sum` meets the specification.
 - (c.) The function `digit_sum` does not meet the specification because it has syntax error(s).
 - (d.) The function `digit_sum` does not meet the specification because it returns incorrect results for all valid argument values.
 - (e.) The function `digit_sum` does not meet the specification because it returns correct results for some but not all valid argument values.
13. We specify that the function `leap`, when applied to any positive integer argument `y` representing a year, should return `True` if the year is a leap year, and `False` otherwise.

```
def leap(y):  
    return (  
        True  
        if y % 400 == 0  
        else True  
        if y % 100 == 0  
        else y % 4 == 0  
    )
```

Which one of the following statements is correct?

- (a.) The function `leap` does not meet the specification because it causes a run-time error.
- (b.) The function `leap` meets the specification.
- (c.) The function `leap` does not meet the specification because it has syntax error(s).
- (d.) The function `leap` does not meet the specification because it returns incorrect results for all valid argument values.
- (e.) The function `leap` does not meet the specification because it returns correct results for some but not all valid argument values.

14. We specify that the function `ilog2`, when applied to any positive integer argument n (> 0), should return the largest integer k , such that $2^k \leq n$.

```
def ilog2(n):  
    def iter(k, acc):  
        return (  
            acc  
            if k == 1  
            else iter(math.floor(k / 2), acc + 1)  
        )  
    return iter(n, 0)
```

Which one of the following statements is correct?

- (a.) The function `ilog2` does not meet the specification because it causes a run-time error.
- (b.) The function `ilog2` meets the specification.
- (c.) The function `ilog2` does not meet the specification because it has syntax error(s).
- (d.) The function `ilog2` does not meet the specification because it returns incorrect results for all valid argument values.
- (e.) The function `ilog2` does not meet the specification because it returns correct results for some but not all valid argument values.

Section D

In Questions 15 and 16, the pre-declared `print` function is used in the Python programs. The `print` function prints the value of its input argument to the console. For example, `print(2 * 5)` prints 10.

15. What is the value printed when the following Python `$1` program is evaluated?

```
x = 1  
def f(x):  
    f = lambda x: x + 2  
    return f(x)  
print(f(x))
```

- (a.) 1
- (b.) 2
- (c.) 3
- (d.) 4
- (e.) An error occurs during the evaluation.

16. What is the value printed when the following Python \$1 program is evaluated?

```
x = 0
f = lambda f, x: f(x + 1)
print(f(lambda x: x + 2, 3))
```

- (a.) 3
- (b.) 4
- (c.) 5
- (d.) 6
- (e.) An error occurs during the evaluation.

17. Consider the following three separate Python \$1 programs.

```
# Program 1
f = lambda x: f(x)
f(1)

# Program 2
f = lambda x: f(x)
f(1)(2)

# Program 3
f = lambda x: f(x)
f(1)(2)(3)
```

Which one of the following statements is correct?

- (a.) All three programs terminate.
- (b.) Program 1 and Program 2 terminate, but Program 3 does not terminate.
- (c.) Program 1 terminates, but Programs 2 and Program 3 do not terminate.
- (d.) None of the three programs terminate.
- (e.) Program 1 and Program 3 terminate, but Program 2 does not terminate.

18. Consider the following three separate Python programs.

```
# Program 1
f = lambda g: g(f)
f(lambda g: f)

# Program 2
f = lambda g: g(f)
f(lambda g: f)(f)

# Program 3
f = lambda g: g(f)
f(lambda g: f)(f)(f)
```

Which one of the following statements is correct?

- (a.) All three programs terminate.
- (b.) Program 1 and Program 2 terminate, but Program 3 does not terminate.
- (c.) Program 1 terminates, but Programs 2 and Program 3 do not terminate.
- (d.) None of the three programs terminate.
- (e.) Program 1 and Program 3 terminate, but Program 2 does not terminate.

END OF PAPER